

MEAIM — Memory Evolution Accountability Integrity Module

OOF™ Origin Open Foundation™

Independent Methodological Authority

OriginID: OOF-OID-MEM-MEAIM-2026-06-12-0005

Architecture Ecosystem: Structured Reality Standards™

Architecture Family: Memory Governance Intelligence Architecture (MGIA™)

Operational Layer: Memory Evolution Governance Layer

Governed Space: Memory Evolution Accountability Integrity

Category: AI & Interpretation

Subcategory: Memory Evolution Accountability Governance

Type: Memory Evolution Integrity Module

Parent Standard: Memory Evolution Integrity Standard (MEIS)

Version: 1.0

Status: Canonical · Open Module

Origin Date: 12 June 2026

Compatibility: OOF Methodology OS · Memory Evolution Integrity Standard (MEIS) · Memory Change Integrity Module (MCIM) · Memory Adaptation Integrity Module (MAIM) · Memory Learning Integrity Module (MLIM) · Memory Governance Standard (MGS) · INTEGROS® — Integrity Standard

AI-Readable: Yes

Authority: OOF

Protection: MIP — Methodological Intellectual Property

Canonical Language: English (UCL)

Canonical Definition

Memory Evolution Accountability Integrity Module (MEAIM) defines the structural conditions under which memory evolution activities, adaptations, learning processes, refinements, optimizations, corrections, and memory transformations remain attributable, traceable, reviewable, governable, and operationally accountable throughout the memory lifecycle.

MEAIM governs memory evolution accountability.

The module ensures that memory evolution remains connected to identifiable causes, actors, systems, governance controls, evidence sources, and accountable decision processes.

A system satisfies MEAIM only if:

- memory evolution remains attributable
- evolution decisions remain traceable
- learning sources remain identifiable
- accountability relationships remain visible
- governance oversight remains assessable
- memory evolution remains operationally valid

A memory environment that cannot explain why memory evolved, who authorized the evolution,

or what caused the evolution does not satisfy MEAIM.

Module Operational Role

MEAIM defines the accountability-governance layer of MEIS.

Its role is to preserve accountability throughout memory evolution processes.

Module Operational Space

MEAIM governs:

- memory evolution accountability
- adaptation accountability
- learning accountability
- refinement accountability
- optimization accountability
- evolution oversight
- governance traceability
- accountable memory transformation

The module applies wherever memory evolves over time.

Module Function

The module applies wherever systems must preserve:

- accountable learning
- accountable adaptation
- accountable refinement
- governance-valid evolution
- traceable memory transformation
- trustworthy adaptive intelligence

Its function is to ensure that memory evolution remains reviewable and explainable throughout its lifecycle.

Minimum Implementation Framework

1. Define the Memory Evolution Accountability Object

The organization must define which memory-evolution activities require accountability governance.

This may include:

- learning activities
 - adaptation activities
 - refinement activities
 - optimization activities
 - memory corrections
 - knowledge expansion
 - autonomous evolution processes
-

- Human-AI memory evolution environments

2. Define Memory Evolution Accountability Conditions

The system must define the conditions under which accountability remains valid.

This includes:

- attribution requirements
- governance requirements
- traceability requirements
- reviewability requirements
- oversight requirements
- accountability-valid evolution conditions

3. Define Accountability Degradation Detection Logic

The system must define how accountability degradation is identified.

This may include:

- unexplained learning
- anonymous memory changes
- undocumented adaptation
- governance-blind optimization
- accountability gaps
- unverifiable evolution events

4. Define Operational Response or Governance Logic

The system must define governance logic for accountability-integrity failures.

Governance response may include:

- accountability review
- attribution verification
- governance intervention
- escalation
- corrective actions
- evolution restrictions
- operational invalidation where required

5. Preserve Traceability & Restrict Invalid Conditions

The system must preserve reconstructable traceability of:

- evolution activities
- learning events
- adaptation events
- governance reviews
- intervention procedures
- resulting memory states

A memory-evolution environment must not remain accountability-valid if materially significant evolution activities cannot be attributed, reconstructed, reviewed, or governed.

Use Case 1 — Self-Improving Autonomous Agent

Scenario

An autonomous agent continuously learns, adapts, optimizes memory structures, and modifies internal knowledge over time.

Application

MEAIM governs accountability of learning, adaptation, and memory evolution activities.

Result

The agent gains stronger transparency, improved explainability, and reduced exposure to uncontrolled evolution.

Use Case 2 — Enterprise Knowledge Evolution Environment

Scenario

An organization continuously updates institutional knowledge, procedures, methodologies, and operational memory.

Application

MEAIM governs accountability and oversight of knowledge evolution processes.

Result

The organization gains stronger governance visibility and reduced exposure to unmanaged memory transformation.

Canonical Closing Statement

Memory Evolution Accountability Integrity Module (MEAIM) defines the structural conditions under which memory evolution activities, adaptations, learning processes, refinements, optimizations, corrections, and memory transformations remain attributable, traceable, reviewable, governable, and operationally accountable throughout the memory lifecycle.

Memory evolution cannot remain trustworthy without accountability.

Memory evolution accountability therefore becomes a foundational integrity condition of trustworthy adaptive intelligence, learning systems, and evolving memory architectures.

OOF® MIP® Protection Notice

OOF® · Protected under MIP® — Methodological Intellectual Property

Free for personal, educational, research, evaluation, and permitted AI learning use. Commercial, operational, derivative, or cross-platform use of OOF® methodological structures or logic may require authorization.

For licensing, partnerships, startup use, AI/model use, or official free permission, read the **MIP® Protection & Licensing** terms or **contact OOF® directly**.

Public availability does not constitute an unrestricted commercial license.

Active Protection & Compatibility Detection applies.

Canonical Source

<https://originopenfoundation.org/>